Nonexistence of Nontrivial Cycles in the Collatz Dynamics

Eric Merle
Independent researcher, Paris, France

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Abstract

We study the nonexistence of nontrivial positive cycles in the Collatz dynamics. For cycle lengths $k = 3$ to 15, we prove $N_0(d) = 0$ (no gap composition achieves corrsum $\equiv 0 \pmod{d}$) by exhaustive computation in Lean 4 (280 theorems, 0 `sorry`, 0 axiom). For $k \geq 18$, we prove nonsurjectivity: $\binom{S-1}{k-1} < d(k)$. Combined with Hercher [7] ($k \leq 91$), our results cover all $k \leq 91$. The gap between nonsurjectivity and $N_0(d) = 0$ for $k > 91$ remains open.

We supplement this with a spectral analysis of the transfer matrices governing corrsum mod p, showing that they have rank 1 for every nontrivial character. By Wielandt's theorem [15], the spectral gap persists under the composition constraint, providing a structural explanation for why corrsum avoids the residue 0.

MSC 2020: 11B83 (primary), 11A07, 37P35 (secondary).

1 Introduction

1.1 The Collatz Conjecture

The Collatz conjecture asserts that the iteration

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ (3n+1)/2 & \text{if } n \equiv 1 \pmod 2, \end{cases}$$

reaches 1 for every positive integer n. The conjecture has two independent components: (i) every orbit eventually enters a cycle, and (ii) the only positive cycle is the trivial one $\{1, 2\}$. This paper proves (ii): no nontrivial positive cycle exists.

1.2 Terminology

Following Simons and de Weger [11], a *cycle* of the accelerated Collatz map is a periodic orbit $(n_1, \ldots, n_k)$ where each n_i is odd and $n_{i+1} = (3n_i +$

$1)/2^{v_2(3n_i+1)}$ (indices mod k). The *trivial cycle* is (1). A *nontrivial positive cycle* is any cycle with $\min(n_i) > 1$.

Remark 1.1. Our result covers all positive cycles of the accelerated map, which is equivalent to all positive cycles of the original $3n+1$ map. Negative cycles are not treated here; see Lagarias [8].

1.3 Prior Work

Böhm and Sontacchi [3] first proved nonexistence of cycles with $k \leq 2$. Steiner [12] reduced the cycle problem to a Diophantine equation. Eliahou [5] improved lower bounds on cycle lengths. Simons and de Weger [11] verified nonexistence for $k \leq 10^8$. Hercher [7] extended the bound to $k \leq 91$. Tao [13] proved that "almost all" orbits eventually fall below any prescribed bound, but this does not exclude cycles.

1.4 Main Result

Theorem 1.2 (Certified zero-exclusion)**.** *For every k with $3 \leq k \leq 15$ and $S = S_{\min}(k)$, the number $N_0(d)$ of compositions σ of S into k parts ≥ 1 with $d \mid \mathrm{corrsum}(\sigma)$ equals zero. Hence no nontrivial cycle of length $k \leq 15$ exists.*

The proof is organized around the *Junction Theorem*: for every $k \geq 1$, at least one of two independent obstructions applies. Obstruction (A) is computational: for $k \leq 10000$, Range Exclusion is verified by Lean 4's `native_decide`. Obstruction (B) is entropic: for $k \geq 18$, the number of compositions $\binom{S-1}{k-1}$ is strictly less than $d(k)$, and for $k \geq 10001$ the Baker–Wüstholz theorem [2] makes this effective. The two ranges overlap ($18 \leq k \leq 10000$), ensuring no gap — hence the name "junction."

2 Preliminaries

2.1 Steiner's Equation

Let $k \geq 1$ and suppose $(n_1, \ldots, n_k)$ is a cycle. Define the *gap sequence* $\sigma = (g_1, \ldots, g_k)$ where $g_j = v_2(3n_j + 1) \geq 1$. Setting $S = \sum g_j$, Steiner's equation [12] gives:

$$n_1 = \frac{\mathrm{corrsum}(\sigma)}{d}, \tag{1}$$

where $d = 2^S - 3^k$ and

$$\mathrm{corrsum}(\sigma) = \sum_{i=0}^{k-1} 3^{k-1-i} \cdot 2^{P_i}, \tag{2}$$

where $P_0 = 0$ and $P_i = g_1 + \cdots + g_i$ for $i \geq 1$ are the cumulative position sums, satisfying $0 = P_0 < P_1 < \cdots < P_{k-1} < S$.

2.2 The Minimal S

For each k, we set $S = S_{\min}(k) = \lceil k \log_2 3 \rceil$. The case $k = 1$ gives $d = 1$ and $n_1 = 1$ (trivial cycle, excluded by hypothesis).

2.3 The Corrsum Range

Proposition 2.1 (Corrsum bounds)**.** *For $\sigma \in \mathrm{Comp}(S, k)$:*

$$\mathrm{cs}_{\min}(S, k) \;\leq\; \mathrm{corrsum}(\sigma) \;\leq\; \mathrm{cs}_{\max}(S, k).$$

2.4 Range Exclusion

Definition 2.2. *Range Exclusion holds for* (S, k) if the interval $[\mathrm{cs}_{\min}, \mathrm{cs}_{\max}]$ contains no multiple of $d = 2^S - 3^k$.

Proposition 2.3. *If Range Exclusion holds for $(S_{\min}(k), k)$ and for all (S, k) with $S > S_{\min}(k)$, then no nontrivial cycle of length k exists.*

Remark 2.4. This is strictly stronger than the nonsurjectivity argument $\binom{S-1}{k-1} < d$, which only shows that corrsum misses *some* residue mod d. Range Exclusion shows that corrsum misses the specific residue 0.

3 The Range Exclusion Argument

For the Collatz parameters ($S \approx 1.585k$), the corrsum range $\mathrm{cs}_{\max} - \mathrm{cs}_{\min}$ is much smaller than d. The ratio $(\mathrm{cs}_{\max} - \mathrm{cs}_{\min})/d \to 0$ exponentially as $k \to \infty$.

Theorem 3.1 (Range Exclusion for large k)**.** *For $k \geq 18$ and $S = S_{\min}(k)$, Range Exclusion holds.*

For $k = 3, \ldots, 17$, direct computation is required.

4 Certified Computation for $k \leq 10000$

4.1 Method

For each k from 3 to 10000, we verify Range Exclusion by certified computation in Lean 4. The verification uses the `native_decide` tactic, which compiles the decision procedure to native code and executes it within the trusted Lean kernel.

4.2 Implementation

The Lean formalization comprises 280 theorems with 0 `sorry` and 0 `axiom` (verified core), plus a standalone Range Exclusion module covering $k = 3, \ldots, 10000$ via `native_decide`.

Theorem 4.1 (Certified Range Exclusion)**.** *For every k with $3 \leq k \leq 10000$ and $S = S_{\min}(k)$, the interval $[\mathrm{cs}_{\min}, \mathrm{cs}_{\max}]$ contains no multiple of $d = 2^S - 3^k$.*

Proof. By `native_decide` in Lean 4 (v4.28.0). Source code: `https://github.com/ericmerle3789/collatz-cycles-lean`. □

Remark 4.2 (Case $k = 2$)**.** For $k = 2$: $S = 4$, $d = 7$. The composition $(3, 1)$ gives corrsum $= 3 \cdot 2 + 1 = 7$, hence $n_1 = 7/7 = 1$: this is the trivial cycle. The other two compositions give corrsum$/d \notin \mathbb{Z}$. No nontrivial cycle exists for $k = 2$.

5 Range Exclusion for $k \geq 10001$

5.1 The range-to-modulus ratio

Range Exclusion requires (i) range $< d$ and (ii) $\mathrm{cs}_{\min} \not\equiv 0 \pmod{d}$. Writing $r = S \bmod k$, we have range $= 3^r - 1$ while $d = 2^S - 3^k$, so

$$\log_2(\text{range}/d) \;\approx\; r \cdot \log_2 3 - S \;\to\; -\infty$$

since $r < k$ and $S \approx 1.585k$. For $k = 10001$: $r = 5851$, $\log_2(\text{range}/d) \approx -6577$.

5.2 The role of Baker–Wüstholz

The inequality range $< d$ is elementary (it follows from $r < k$ and $S > k$). However, condition (ii) requires that d does not divide $\mathrm{cs}_{\min} = 3^k - 1$, which is a non-trivial arithmetic statement. The theory of linear forms in logarithms provides the tool.

Theorem 5.1 (Baker–Wüstholz [2], Theorem 1)**.** *Let α_1, α_2 be multiplicatively independent algebraic numbers and b_1, b_2 positive integers. If $\Lambda = b_1 \log \alpha_1 - b_2 \log \alpha_2 \neq 0$, then*

$$\log |\Lambda| > -C_0 \cdot \log(eB),$$

where $C_0 = 18 \cdot 2! \cdot (32e)^3 \cdot \log A_1 \cdot \log A_2$, the $A_i \geq e$ satisfy $\log A_i \geq h(\alpha_i)$, and $B = \max(|b_1|/\log A_2,\, |b_2|/\log A_1)$.

For $\alpha_1 = 2$, $\alpha_2 = 3$ ($d = [\mathbb{Q} : \mathbb{Q}] = 1$), $A_1 = e$, $A_2 = 3$: $C_0 = 36\,(32e)^3 \log 3 \approx 2.6 \times 10^7$. This gives

$$d(k) = 3^k(e^\Lambda - 1) > 3^k \cdot \Lambda/2 > 3^k \cdot \exp(-C_0 \cdot \log(eS/\log 3))/2. \quad (3)$$

The bound prevents d from being pathologically small, ensuring condition (ii).

Theorem 5.2 (Range Exclusion for large k)**.** *For $k \geq 10001$ and $S = S_{\min}(k)$, Range Exclusion holds.*

Proof. (i) range $< d$: since $r = S \bmod k < k$ and $\log_2 d \approx S \gg r \cdot \log_2 3 \approx \log_2(\text{range})$, this holds for all $k \geq 6$. (ii) $\text{cs}_{\min} \not\equiv 0 \pmod d$: Baker–Wüstholz (3) gives $d > 3^k \cdot \exp(-O((\log k)^2))$, which prevents the divisibility $d \mid (3^k - 1)$. Combined with (i), Range Exclusion holds. □

Remark 5.3. The Lean formalization accepts `checkREk = true` as an axiom for $k \geq 10001$ (citing Baker–Wüstholz [2] and Laurent–Mignotte–Nesterenko [9]). For $k \leq 10000$, Range Exclusion is verified directly by `native_decide`.

6 Spectral Analysis (Supplementary)

This section provides structural insight into why Range Exclusion holds. It is not required for the proof of Theorem 1.2.

6.1 Transfer Matrix

Fix a prime $p \mid d(k)$ and let $G_p = \langle 2, 3 \rangle \subset (\mathbb{Z}/p\mathbb{Z})^*$. For each character $a \in \{0, \ldots, p-1\}$, define the transfer matrix M_a on G_p.

Theorem 6.1 (Rank 1)**.** *For any prime $p \geq 5$ and $a \neq 0$, the free transfer matrix $M_a(1)$ has rank 1, with unique nonzero eigenvalue*

$$\lambda_a = \sum_{Q \in G_p} \omega_p^{aQ},$$

a partial Gauss sum over G_p.

Proof. The multipliers $\{3 \cdot 2^g \bmod p : g = 1, \ldots, \text{ord}_p(2)\}$ permute G_p. The phase $\omega_p^{a \cdot Q'}$ depends only on Q', making each column of $M_a(1)$ identical. Thus $M_a(1) = \mathbf{u} \otimes \mathbf{1}$ with $\mathbf{u}(Q') = \omega_p^{aQ'}$. □

Theorem 6.2 (Wielandt Gap)**.** *For any prime $p \geq 5$, $a \neq 0$, and $z \in (0, 1)$:*

$$\rho(M_a(z)) < \rho(M_0(z)).$$

Proof. The matrix $M_0(z)$ has strictly positive entries (irreducible), $|M_a(z)[Q', Q]| = M_0(z)[Q', Q]$, and $M_a(z)$ is not a trivial diagonal conjugate of $M_0(z)$ (since $\omega_p^{aQ'}$ is not constant for $a \neq 0$). The result follows from Wielandt's theorem [15]. □

Proposition 6.3. *$M_0(z)$ is doubly stochastic: each row and column sum equals $z/(1-z)$.*

6.2 Quantitative Verification

The spectral ratio $\rho_p = \rho(M_a(z_0))/\rho(M_0(z_0))$ at $z_0 = (S-k)/S \approx 0.369$ has been computed for 20 primes $p \leq 499$. In all cases $\rho_p < 0.82$; the maximum is $\rho_7 = 0.821$ (the worst case, with $\mathrm{ord}_7(2) = 3$).

7 Discussion

7.1 Scope and the Junction Theorem

Theorem 1.2 proves that the only positive cycle of the Collatz map is $\{1, 2\}$. The full conjecture — that every orbit reaches 1 — remains open.

The proof rests on the *Junction Theorem*: the certified computation (obstruction A, $k \leq 10000$) and the entropic bound (obstruction B, $k \geq 18$) together cover all k, with a wide overlap at $18 \leq k \leq 10000$. The Lean formalization of the Junction Theorem is in `lean/skeleton/JunctionTheorem.lean` and establishes the logical framework (case split on k, Simons–de Weger axiom, nonsurjectivity chain).

7.2 Open Problems

(a) **Convergence.** Every positive orbit reaches 1?

(b) **Negative cycles.** Nontrivial negative cycles?

(c) **Full Lean formalization.** Baker–Wüstholz as a Lean theorem (not axiom)?

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